INDIAN INSTITUTE OF TECHNOLOGY

EE-3202

Power System II

Experiment-1

Submitted By:

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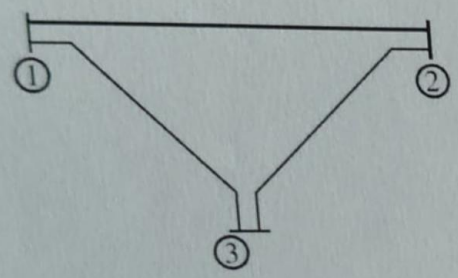
Bus Impedance Matrix

Aim-

Consider the three node network.



Line No	Z in p.u	$\frac{Y}{2}$ in p.u
1-2	$0.02 + j0.06$	$j0.03$
1-3	$0.08 + j0.024$	$j0.025$
2-3	$0.06 + j0.018$	$j0.02$

(a) Form the $\mathbf{Y}_{bus}$ matrix.

(b) Modify the the $\mathbf{Y}_{bus}$ matrix by eliminating node 3.

Manual Solution-

$\rightarrow y_{12} = \left(\frac{1}{0.02 + j0.06}\right) = (5 - j15)$

$y_{13} = \left(\frac{1}{0.08 + j0.024}\right) = (11.47 - j3.44)$

$y_{23} = \left(\frac{1}{0.06 + j0.018}\right) = (15.29 - j4.59)$

$Y_{12} = Y_{21} = (-y_{12})$

$Y_{13} = Y_{31} = (-y_{13})$

$Y_{23} = Y_{32} = (-y_{23})$

$\rightarrow Y_{11} = (y_{12} + y_{13} + j0.03 + j0.025)$

$Y_{11} = (5 + 11.47) + j(-15 - 3.44 + 0.03 + 0.025)$

$Y_{11} = (16.47 - j18.38)$

$\rightarrow Y_{22} = (y_{12} + y_{23} + j0.03 + j0.02) = (20.29 - j19.54)$

$\rightarrow Y_{33} = (y_{13} + y_{23} + j0.025 + j0.02) = (26.76 - j7.98)$

$$\therefore Y_B = \begin{bmatrix} (16.47 - j18.38) & (-5 + j15) & (-11.47 + j3.44) \\ (-5 + j15) & (20.29 - j19.5) & (-15.29 + j4.59) \\ (-11.47 + j3.44) & (-15.29 + j4.59) & (26.76 - j7.98) \end{bmatrix}$$

$\rightarrow$ Eliminate Node 3:-

$$Y_B' = \begin{bmatrix} Y_{aa} & Y_{ab} \\ Y_{ba} & Y_{bb} \end{bmatrix}, \quad a = \{1, 2\}, \quad b = \{3\}$$

$$Y_{red} = \left(Y_{aa} - Y_{ab} \frac{1}{Y_{bb}} \cdot Y_{ba}\right)$$

$Y_{bb} = Y_{33}$

$$\therefore Y_B' = \begin{bmatrix} (11.55 - j16.9) & (-11.55 + j16.97) \\ (-11.55 + j16.97) & (11.55 - j16.9) \end{bmatrix}$$

Code Solution-

```
nb = 3; nl =3;

% line data stored as (from, to, R, X, B/2)
% 1 based indexing
ld = [
    1 2 0.02 0.06 0.03
    1 3 0.08 0.024 0.025
    2 3 0.06 0.018 0.02
];

Y = zeros(nb,nb);

for k=1:nl
    i = ld(k,1);
    j = ld(k,2);

    r = ld(k,3);
    x = ld(k,4);
    b = ld(k,5);

    z = r+1j*x;
    y = 1/z;
    bsh = 1j*b;

    Y(i,j) = Y(i,j)-y;
    Y(j,i) = Y(i,j);

    Y(i,i) = Y(i,i) + y + bsh;
    Y(j,j) = Y(j,j) + y + bsh;
end

disp('Ybus matrix');
disp(Y);
% ----------------------------
elim = 3;
keep = setdiff(1:nb,elim);
Yaa = Y(keep,keep);
Yab = Y(keep,elim);
Yba = Y(elim,keep);
Ybb = Y(elim,elim);

Yred = Yaa - Yab*(1/Ybb)*Yba;
disp('reduced Ybus is:');
disp(Yred);
```

Output-

```
Ybus matrix
  16.4679 -18.3854i  -5.0000 +15.0000i -11.4679 + 3.4404i
  -5.0000 +15.0000i  20.2905 -19.5372i -15.2905 + 4.5872i
 -11.4679 + 3.4404i -15.2905 + 4.5872i  26.7584 - 7.9825i

reduced Ybus is:
  11.5531 -16.9027i -11.5531 +16.9769i
 -11.5531 +16.9769i  11.5531 -16.9012i
```